Predicting Diabetic Distress and Emotional Burden in Type-2 Diabetes Using Explainable AI

Understanding the Unseen: Emotional Burden in Type-2 Diabetes

PROJECT OVERVIEW

The Problem

Many Type-2 diabetic patients experience significant, yet often hidden, emotional distress and psychological burden, impacting their overall health and treatment adherence.

Our Objective

Develop an Explainable AI (XAI) system to accurately predict diabetic distress and emotional burden, offering transparent insights into these complex conditions.

Key Tools

Leveraging Python, Django/Flask, XGBoost, SHAP, SQLite, and HTML/CSS for a robust and user-friendly solution.

Expected Outcome

A clear prediction of distress levels (Low, Moderate, High) with actionable, transparent explanations for both clinicians and patients, facilitating early intervention.

PROJECT OUTLINE

A Structured Approach to Emotional Well-being Prediction

01

Modular Design

The system comprises Admin and User modules. Admin manages users, complaints, and reviews, while Users handle signup, login, predictions, profile management, and result viewing.

02

AI Integration Pipeline

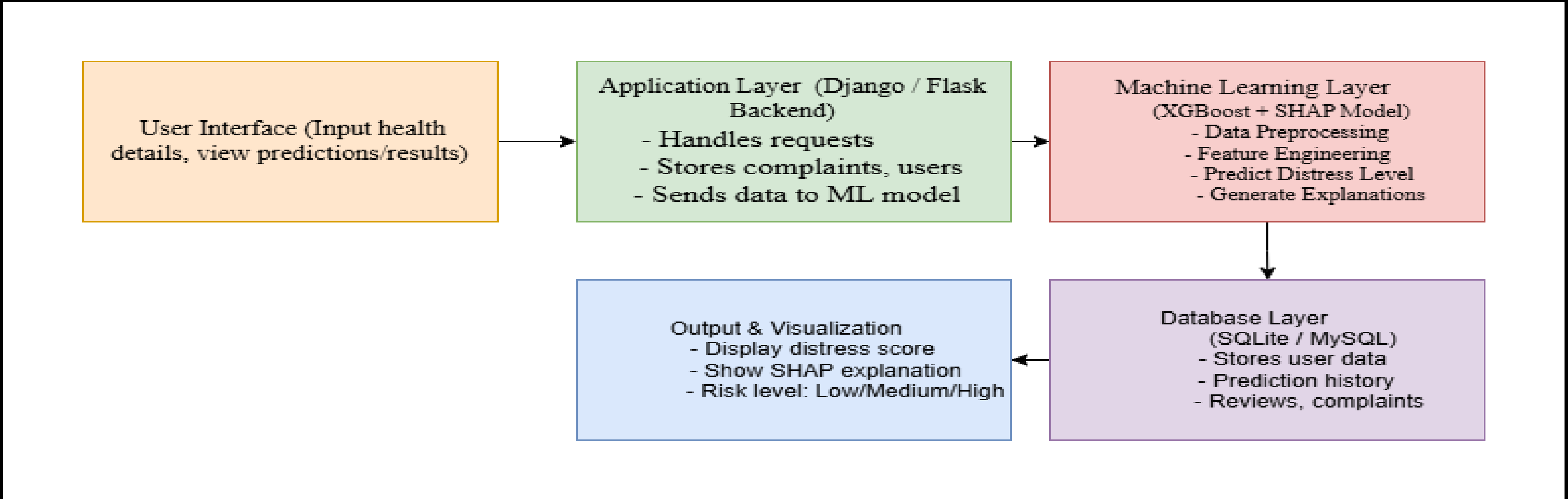
Our AI core integrates dataset preprocessing, an XGBoost model for prediction, and SHAP (SHapley Additive exPlanations) for crucial explainability.

03

Predictive Outcome

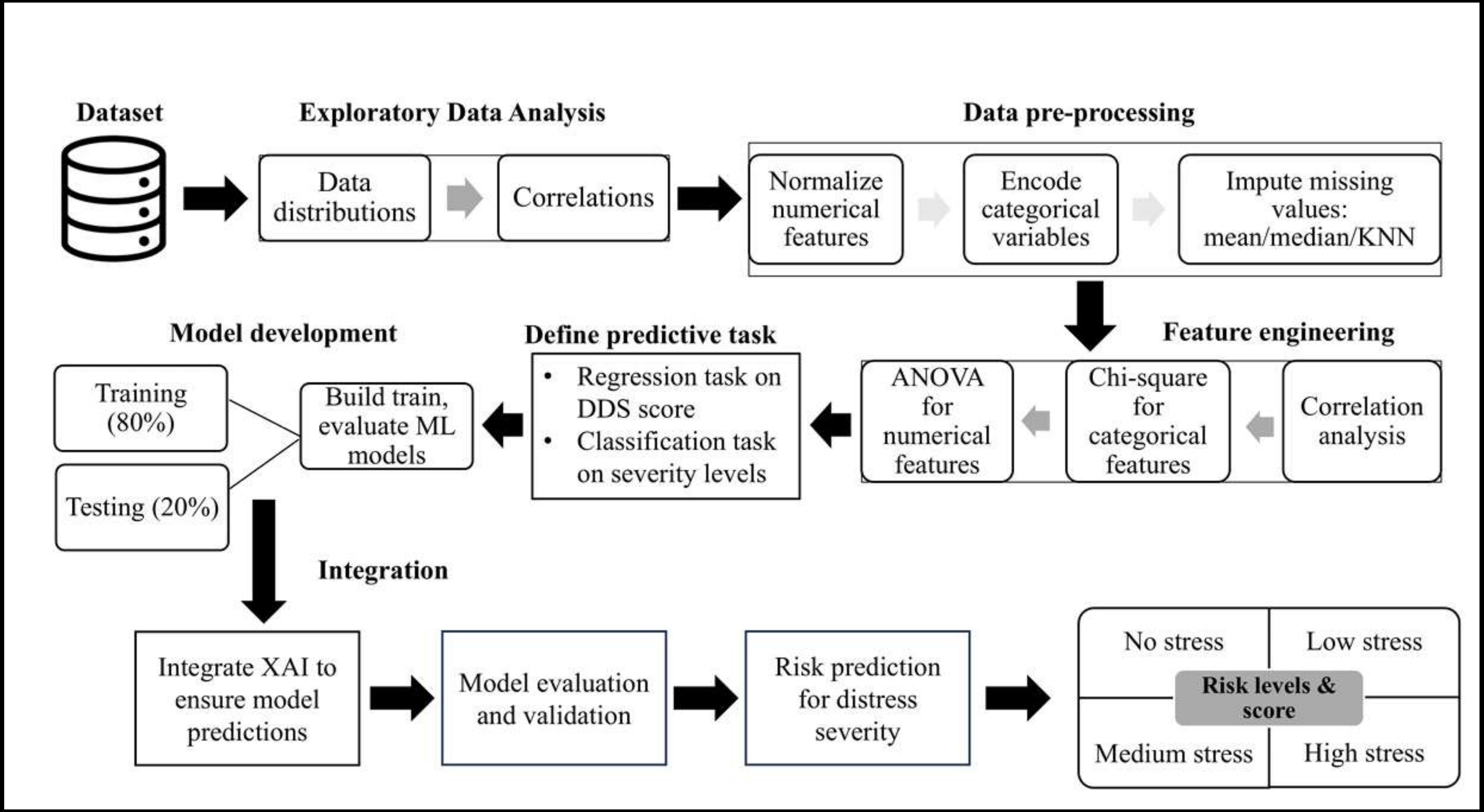
The ultimate goal is to provide a transparent prediction of diabetic distress levels, enabling timely interventions and personalized care.

SYSTEM ARCHITECTURE



User Interface: Collects input like age, glucose level, emotional burden.
Application Layer: Validates user input, manages sessions, and routes data.
Machine Learning Layer: Uses XGBoost model to predict distress and SHAP for explainability.
Database: Stores all user details, complaints, reviews, and predictions.
Output Layer: Shows results (distress level + explanation graph).

PROCESS FLOW DIAGRAM



TECHNOLOGIES USED

Layer	Technology
Frontend	HTML, CSS, Bootstrap
Backend	Django / Flask
ML Framework	XGBoost, scikit-learn, SHAP
Database	SQLite / MySQL
Visualization	Matplotlib / Plotly

SPRINT BACKLOG

SPRINT1

Backlog term	Date	Original Estimation (hrs)	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7	Day 8	Day 9	Day 10
Project setup & environment configuration	29/12/25	3	1	1	1	0	0	0	0	0	0	0
Design database schema (users, complaints, predictions)	31/12/25	3	1	1	1	0	0	0	0	0	0	0
Admin & User authentication (login/signup)	03/01/26	3	1	1	1	0	0	0	0	0	0	0
Frontend (UI design for Home, Login, Signup)	06/01/26	3	1	1	1	0	0	0	0	0	0	0
Session management & testing	09/01/26	2	1	1	0	0	0	0	0	0	0	0
		14	5	5	4	0	0	0	0	0	0	0

SPRINT2

Backlog term	Date	Original Estimation (hrs)	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7	Day 8	Day 9	Day 10
Admin dashboard (view/manage users)	19/01/26	3	1	1	1	0	0	0	0	0	0	0
Change password module (Admin)	22/01/26	3	1	1	1	0	0	0	0	0	0	0
Complaint & review submission (User)	25/01/26	3	1	1	1	0	0	0	0	0	0	0
User profile management	28/01/26	3	1	1	1	0	0	0	0	0	0	0
Testing and integration (Admin + User dashboard)	31/01/26	2	1	1	0	0	0	0	0	0	0	0
		14	5	5	4	0	0	0	0	0	0	0

SPRINT3

Backlog term	Date	Original Estimation (hrs)	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7	Day 8	Day 9	Day 10
Dataset collection & preprocessing	09/02/26	3	1	1	1	0	0	0	0	0	0	0
Train XGBoost model for prediction	12/02/26	3	1	1	1	0	0	0	0	0	0	0
Integrate SHAP for explainability	15/02/26	3	1	1	1	0	0	0	0	0	0	0
Develop API for AI prediction	18/02/26	3	1	1	1	0	0	0	0	0	0	0
Display results & graphs in UI	21/02/26	2	1	1	0	0	0	0	0	0	0	0
		14	5	5	4	0	0	0	0	0	0	0

SPRINT 4

Backlog term	Date	Original Estimation (hrs)	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7	Day 8	Day 9	Day 10
Testing of AI model & performance tuning	01/03/26	3	1	1	1	0	0	0	0	0	0	0
Visualization of past results & reports	05/03/26	3	1	1	1	0	0	0	0	0	0	0
Final UI polish and validation	08/03/26	3	1	1	1	0	0	0	0	0	0	0
Documentation & presentation prep	12/03/26	2	1	1	0	0	0	0	0	0	0	0
Deployment (local/cloud) and demo	16/03/26	3	1	1	1	0	0	0	0	0	0	0
		14	5	5	4	0	0	0	0	0	0	0

THANK YOU